Python DA Assignment 2 – Data Visualization

Taxi Trip Data Analysis

Project Overview

This project analyzes the Seaborn **Taxis dataset** using Python, Pandas, Matplotlib, and Seaborn. The objective was to clean the taxi trip data, handle missing values, explore important variables, and create visualizations to understand fare, distance, tips, payment methods, and pickup patterns.

Data Cleaning and Preparation

The dataset contains **6,433 taxi trip records and 15 columns**. Missing values were identified in the payment, pickup_zone, dropoff_zone, pickup_borough, and dropoff_borough columns. Since these were categorical columns, missing values were handled using the **mode** of the respective columns. After imputation, the dataset was checked again to ensure that the missing values were handled successfully.

The pickup column was converted into **datetime format** to support time-based analysis.

Key Findings

- The **line chart** shows how taxi fares vary over different pickup times.
- The **bar chart** compares the total fare generated from different pickup boroughs and helps identify the borough contributing the highest total fare.
- The **pie chart** shows the distribution of trips according to payment method, allowing comparison between credit card, cash, and other payment types.
- The **histogram** shows the distribution of taxi trip distances and helps identify the most common trip-distance range.
- The **box plot** compares tip amounts across pickup boroughs and helps identify variations and potential outliers.
- The **count plot** shows the number of trips originating from each pickup borough.
- The **scatter plot** demonstrates the relationship between trip distance and fare. It helps determine whether longer trips generally result in higher fares.
- The **heatmap** highlights correlations between numerical variables such as distance, fare, tip, tolls, and total.
- The **pair plot** provides a visual comparison of relationships among distance, fare, tip, and total across different pickup zones.

- The **violin plot** compares fare distributions across different payment methods and shows differences in fare spread.

Conclusion

The analysis demonstrates how data visualization can be used to understand taxi trip behavior and identify meaningful patterns in transportation data. By combining data cleaning with Matplotlib and Seaborn visualizations, the project provides insights into **fare patterns, trip distance, payment methods, tips, pickup locations, and relationships between numerical variables**. These insights can help taxi service providers better understand customer behavior and improve operational decision-making.

Tools and Technologies

Python | Pandas | NumPy | Matplotlib | Seaborn | Google Colab

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